

Qiushi Guo

Computer Vision Researcher | Robust and Data Efficient Visual Learning

+86 189 1916 0426 | guoqiushi910@gmail.com | [Google Scholar](#) | [GitHub](#)

RESEARCH PROFILE

Computer vision researcher and engineer with research and deployment experience across intelligent vehicles, railway safety, and financial AI. Research focuses on robust and data-efficient visual learning under domain shift, degraded data, occlusion, and limited supervision, with work spanning face analysis, object detection, semantic segmentation, synthetic data generation, and RGB/IR perception. Experienced in translating real-world failure modes into research questions, building datasets and evaluation protocols, and deploying optimized models in safety-critical systems.

RESEARCH INTERESTS

Robust visual recognition; data-efficient learning; domain adaptation and generalization; object detection and semantic segmentation; face detection, recognition, quality assessment, and anti-spoofing; synthetic data and data augmentation; multimodal RGB/IR perception.

EDUCATION

M.Sc. in Computer Science, Specialization in Artificial Intelligence

Technische Universität Kaiserslautern and German Research Center for Artificial Intelligence DFKI, Germany

2017 - 2020

- Master's thesis: Calibration-Free Induced Magnetic Field Indoor and Outdoor Positioning Via Data-Driven Modeling.

B.Eng. in Computer Science and Technology

University of Electronic Science and Technology of China UESTC, Chengdu, China

2012 - 2016

RESEARCH AND PROFESSIONAL EXPERIENCE

Senior Algorithm Engineer

Great Wall Motor Company Limited

Sep 2025 - Present

- Research and develop face detection, face recognition, and face image quality assessment for driver and occupant monitoring systems under illumination changes, blur, occlusion, pose variation, and complex cabin backgrounds.
- Designed progressive curriculum learning and ghost-aware heatmap supervision to improve real-world face detection stability and suppress false positives; the resulting work was accepted at FG 2026.
- Developed and quantized production models for vehicle deployment, supporting DDAW certification testing and ongoing Euro NCAP-related evaluation.
- Led the end-to-end OMS Face-ID pipeline, including data specifications, model training and quantization, face detection, quality-based filtering, enrollment, matching, and open-set and closed-set evaluation.

Senior Algorithm Engineer

CSRD

May 2024 - Sep 2025

- Developed robust railway obstacle intrusion detection for adverse weather, long-range targets, small objects, and low-error-tolerance safety scenarios.
- Proposed an optical-flow-guided semi-supervised segmentation approach that reframed obstacle detection as track-region intrusion perception and improved event-level reliability.
- Built and released SynRailObs, a controllable synthetic dataset covering people, vehicles, animals, rocks, motorcycles, weather conditions, distances, and railway scenes.

RESEARCH AND PROFESSIONAL EXPERIENCE CONTINUED

Algorithm Engineer, Artificial Intelligence Laboratory

Mar 2021 - Mar 2024

China Merchants Bank

- Improved face anti-spoofing against replay, print, and cut-out attacks in financial identity verification and deployed the resulting models in production.
- Developed face occlusion detection and quality-control models for robust face-recognition pipelines.
- Contributed to a real-time food tray recognition and auto-billing system based on instance segmentation.

Research Intern

2019

German Research Center for Artificial Intelligence DFKI

- Conducted multi-scale machine learning research for building property value classification and contributed to the resulting publication at IGARSS 2019.
- Completed the master’s thesis [Calibration-Free Induced Magnetic Field Indoor and Outdoor Positioning Via Data-Driven Modeling](#), developing a data-driven approach to calibration-free indoor and outdoor positioning.

SELECTED PUBLICATIONS

- **DmsFIQA: DMS-Specific Face Image Quality Assessment for In-Cabin Driver Monitoring System.** CVPR 2026 Biometrics Workshop, oral presentation. *Qiushi Guo, Zhiye Lin, Yuanqing Luo, Jason Rambach*
- **Progressive Curriculum Learning and Ghost-Aware Supervision for Real-World Face Detection.** IEEE International Conference on Automatic Face and Gesture Recognition FG 2026. *Qiushi Guo, Zhiye Lin, Yuanqing Luo*
- **A Universal Railway Obstacle Detection System Based on Optical-Flow Guided Semi-Supervised Segmentation.** VEHITS, Best Poster Paper Nomination. *Qiushi Guo, B. Cao, D. Hao, C. Wang, L. Chen, P. Yan*
- **SynRailObs: A Synthetic Dataset for Obstacle Detection in Railway Scenarios.** Under review; dataset available on Kaggle and GitHub. *Qiushi Guo, Jason Rambach*
- **CACP: Context-Aware Copy-Paste to Enrich Image Content for Data Augmentation.** CVPR Workshop on Exploring the Next Generation of Data. *Qiushi Guo, S. Wang, C. Chang, J. Rambach*
- **A Real-Time Chinese Food Auto-Billing System Based on Instance Segmentation.** 2023 IEEE Region 10 Symposium (TENSYP), 1-5. *Qiushi Guo, Yifan Chen, Yihang Yao, Tengting Zhang, Jin Ma*
- **Multi-Scale Machine Learning for the Classification of Building Property Value.** IEEE International Geoscience and Remote Sensing Symposium IGARSS 2019. *P. Helber, B. Bischke, Q. Guo, J. Hees, A. Dengel*

RESEARCH COLLABORATION AND OPEN SCIENCE

- Maintain an active research collaboration with Dr. Jason Rambach at DFKI on synthetic data, robust visual perception, and face image quality assessment for in-cabin monitoring.
- Released the SynRailObs dataset and supporting resources through GitHub and Kaggle to enable reproducible research on railway obstacle detection under diverse conditions.

TECHNICAL SKILLS

Programming and tools: Python, C/C++, Linux, Git, Docker, LaTeX.

Frameworks and deployment: PyTorch, TorchVision, OpenCV, ONNX, TensorRT, model quantization.

Languages: English professional working proficiency; German working proficiency; Chinese native.